YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 6 OF 10 · 40 MINUTES

Footprints and height 脚印与身高

Linked measurements (a correlation) let a scatter graph and line of best fit estimate one from the other; people vary, so give a range.

- I can measure foot length and height, plot a scatter graph and draw a line of best fit.
- I can estimate the footprint owner's height as a range from my graph and name suspects who fit.

foot length 脚长 · **height** 身高 · **data** 数据 · **scatter graph** 散点图 · **correlation** 相关性 · **line of best fit** 最佳拟合线 · **estimate** 估计 · **range** 范围

THE LESSON

Do Now	4	New case leader intro; 3 Week 5 recap questions from the evidence log; Parking Board; goals
Hook	3	True-size footprint by a metre stick: can it tell height? Vote short, medium or tall
Learn	7	Scatter graph, correlation, line of best fit; read off a range; model measuring safely, kindly
Do	18	Measure in pairs, class data table, plot scatter graph, line of best fit, read off at 24 cm
Think and Reflect	8	Which suspects fit the range? Pair talk; evidence log row 6, tracker; exit ticket with vote

Hands-on: Pairs measure foot length (shoes off, 0.5 cm) and height (1 cm), plot a scatter graph and read off at 24 cm.

Collect or assess: Scatter graph and estimate (rubric strand 4); evidence log row 6, tracker, W6 confidence; exit ticket with Week 7 vote

SAFETY

- Shoes off, socks on: shoes under desks, back on before anyone leaves their seat
- No standing on chairs or tables; never swing metre sticks; retract tapes slowly
- Measuring is voluntary: numbers not names, no comments on size; sample data allowed

WATCH OUT

- Measuring slow or under 16 measured: switch all to the sample data card; never mix data on one graph.
- Joining dots, starting at zero or one number: ask 'As many crosses above as below?' Insist on a range.

Language: Frame: 'The person was probably between ___ cm and ___ cm tall.' Say 'height', not 'high'. Chinese answers count.

Next week: Exit tickets: give each group one option (Soil, Paper, Ink 2, Powder 2), sets Week 8 kit; collect Week 7 soil

PREP FOR ONE CLASS

before the lesson about 103 min (43 could be done by someone else); after about 28 min (some times estimated)

THE WEEK BEFORE

- [ ] Meet current teacher: handover, tracker, freezer decision, Weeks 6, 7 kit 30 min · teacher
- [ ] Read Case Bible Sections 4 to 6 (suspects' heights and foot lengths) 20 min · teacher
- [ ] Send print job (rebuilt designs): worksheets, data cards, signs, kit card 5 min · reprographics
- [ ] Print footprint at 100% (not fit to page): 6 per class plus 1; check 24 cm 5 min · technician
- [ ] Laminate the 7 sample data cards in A5 pouches (optional) optional 10 min · technician
- [ ] Borrow 6 hardbacks; collect rulers, metre sticks, trays; sharpen 24 pencils 20 min · technician
- [ ] Find the Week 5 Ideas Parking Board or its photo for the Do Now 5 min · teacher

THE DAY BEFORE

- [ ] Set up 2 or 3 height stations (chart or metre stick, sign); check the scale 8 min · technician
- [ ] Check the slide 9 data table can be typed into while projected 3 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Decide: class's own data or sample data (sample if class small) 2 min · teacher
- [ ] Set out group trays, case files, role and frame cards; Hook kit at front 5 min · technician

AFTER THE LESSON

- [ ] Clean-up: shoes on, kit counted, data cards collected, footprints filed 3 min · students
- [ ] Read exit tickets: record Week 7 option choices, give each group one option 15 min · teacher
- [ ] Last class: return kit and books, store data cards; leave charts if needed 10 min · technician

PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Scene footprint x6 per class, A4 at 100%, check 24 cm (1 per group)
- Sample class data card x6, A5, laminated if possible (1 per group)
- Height station sign x3, A4 (1 per station)
- Teacher copies: kit card (A5), sample data card, Hook footprint

GET OR MAKE (NEW THIS LESSON)

- Rulers, 30 cm x12 (2 per group)
- Metre sticks or tape measures x7 (6 groups, 1 Hook)
- Hardback books x6 (borrow from library or staff room)
- Paper height charts, 130 to 190 cm x3 (buy ready-made)
- Masking tape x1 roll
- Up to 3 metre sticks for upright stations (optional)
- Laminating pouches, A5 x7 (optional)
- Make: height stations x3 (chart, masking tape, sign)

Have ready from earlier weeks: Class box, Case files x24, Scientist's Logbooks x24, My Science Dictionaries x24, Role card and frame card sets x6 each, Group trays x6, Sharp pencils x24 (4 per group), Ideas Parking Board (Week 5) or photo